

# Data Literacy for Political Science

POLI-LIT

Chen & Hwang 2025

- This is planned as a mid-semester lecture — use this slide to signal a shift in topic: "Today we're starting something a little different."
- Frame POLI-LIT as a practical, skills-based component of the course — not a detour, but a complement to the substantive content they've been learning.
- Slides by Edwina Chih-Yu Chen & Insu Hwang (Dept of Political Science, USC)
- Supported by the Center for Teaching Excellence (CTE) GTA/TA Teaching Innovation Grant.
- These slides are designed for use with the POLI-LIT: Political Science Data Literacy Module. Content draws from various websites, textbooks, and academic sources, adapted for introductory-level instruction.
- For updates, additional materials, or the latest version of the module, visit: <https://www.edwinachen.com/teaching/poli-lit>

# Outline

- What is POLI-LIT? Why data literacy?
- Political Science Research Methods
- Making Statistical Inference
- Conceptualization to Measurement - V Dem EDI
- POLI LIT Unit 1 Preview

- Walk students through the roadmap briefly so they know what to expect.
  - Emphasize that today's lecture is the only in-person session for POLI-LIT — the rest will be online and self-paced.
  - Suggested phrasing: "We'll cover a lot of ground today, but the goal is to give you a foundation. The online units will let you go at your own pace."
- 
- Instructor context: This lecture was first piloted in Fall 2025 for POLI 101: Introduction to Global Politics at USC, with approximately 80 students. Some foundational concepts covered in this lecture (e.g., correlation vs. causation, IV/DV) were taught earlier in the semester, so parts of this lecture served as review for students. If you are adopting this module, adjust the pacing and depth based on what your students have already encountered in your course.
  - In the pilot, the lecture ran approximately 1 hour 10 minutes. Depending on class size and how much discussion you encourage, it could run longer. You can also split it into two class sessions.
  - In the pilot, we gave students a PDF copy of the slides after the in-person lecture. We chose not to distribute slides beforehand so students would concentrate on the lecture itself. If you prefer not to share the slides, we recommend providing a structured handout or fill-in-the-blank style notes so students have something to follow along with and refer back to.

# What is POLI-LIT?

## Political Data Literacy module

### Edwina Chen and Insu Hwang Political Science

Edwina Chen is a Ph.D. candidate in Political Science. Originally from Taiwan, she earned her bachelor's in political science from National Cheng Kung University (TW) and a master's in politics from the University of Essex (UK). Her research interests include LGBTQ+ politics, queer theory, authoritarianism, democratization, and East Asian politics. Her current work examines global patterns in same-sex marriage legalization and how voters perceive LGBTQ+ political candidates. Outside of her academic work, Chen loves traveling and learning from the cultures and stories she encounters along the way. Professional website: [edwinachen.com](http://edwinachen.com)



Edwina Chen

Insu Hwang is a Ph.D. student in Political Science at USC. He studies varied topics in International Relations and Comparative Politics. He earned his B.A. in International Studies and an M.A. in International Studies at Pukyung National University in Busan, Republic of Korea. His research interest includes the re-balancing of Global Power, particularly focusing on the concept of 'Middle Power' amidst the rise of Indo-Pacific region, as well as Asian security and the role of International Institutions.



Insu Hwang

### Project: R and Data Literacy Module for Political Science Undergraduates

This project will develop a self-paced, web-based instructional module for POLI 101: Introduction to Global Politics. The resource aims to strengthen data literacy by introducing students to political datasets and the basics of R programming, while also teaching usage of artificial intelligence tools such as ChatGPT for learning support. By integrating real-world datasets, the module will help students interpret data structures, understand descriptive statistics, and create simple visualizations. Deliverables will include a structured module (3–4 units), an instructor manual with guidance on course integration, assessments, and a curated resource hub. Although designed for POLI 101, the module will be adaptable for asynchronous or hybrid use in other political science or social science courses.

- (Optional) Briefly acknowledge the project creators (Edwina Chen & Insu Hwang) and grant for building the module.
- Consider replacing the current bio/grant image with a visual that connects to your course content and data literacy more broadly — for example, a screenshot of a political dataset, a V-Dem democracy map, or a data visualization from Our World in Data. This makes the slide more content-forward and connect the module to what students are already studying.

# What is POLI-LIT?

## Political Data Literacy module

Designed to provide you with:

- **Data skills** — Learn how to read and interpret datasets and codebooks
- **R skills** — Use R to organize, summarize, and visualize political data
- **AI literacy** — Use tools like ChatGPT to assist (not replace!) your analysis
- **Critical thinking** — Question how data is constructed and used in political arguments

- Go through each of the four areas briefly. The key message is that these skills build on each other across the three units — data literacy first, then R coding, then AI-assisted approach.
- Reassure students: they are not expected to become data scientists. The goal is comfort and familiarity, not mastery.
- Highlight the "assist, not replace" framing for AI literacy — this sets the tone for Unit 3 and prevents students from thinking AI tools will do the work for them.

# Why Data Literacy Matters in Politics

Many political claims rest on data:

“Crime is rising in urban areas.”    “Voter turnout declined last year.”

“Democracy is eroding globally.”

But

Who collected the data?

How was *crime* or *democracy* defined?

What choices were made in measurement?

- Read one or two of the claims aloud and pause.
- Ask students: "Have you seen claims like these recently? Where — social media, news, a conversation?"
- Then pivot to the "But" questions on the slide. The point is that even familiar-sounding claims depend on decisions about data collection, definition, and presentation.

# Why Data Literacy Matters in Politics

Many political claims rest on data; without data literacy, you consume what is fed to you.

POLI-LIT gives you tools to answer those questions and interpret claims.

We want you not just to consume political data, but to interrogate it.

- This is the motivational core of the module. Drive home the message: data literacy isn't about being skeptical of everything — it's about being equipped to ask better questions, get better answers.
- Suggested phrasing: "We're not trying to make you distrust data. We're trying to make sure you can tell good data from bad data, and strong claims from weak ones."



- Transition: "Before we work with data, we need to understand how political scientists ask and answer questions."

This section draws on content from two textbooks used in building this module:

- Toshkov, Dimitar. Research Design in Political Science. Palgrave Macmillan, 2016.
- Franco, Josh, et al. Introduction to Political Science Research Methods, 1st/2nd Edition. Open Education Resource (CC BY-NC). Free access: <https://ipsrm.com/>

# Types of Research

We ask questions in three broad ways:

## **Descriptive:**

What is happening? → Describe facts, trends, frequencies

“What percent of eligible citizens voted?”

- Introduce the three types of research one at a time across Slides 8–10.
- Start here with descriptive research. Emphasize that this is the most basic and common type — it answers "what is happening?"
- Use the example on the slide and connect it to something students already know from the course, e.g., a reading or discussion about voter turnout.

# Types of Research

We ask questions in three broad ways:

## **Predictive:**

What will happen? → Forecast outcomes

Example: “Which party will win the next election?”

- Explain that predictive research is forward-looking — it tries to forecast what will happen.
- Use the example, note that prediction is common in media (polls, models) etc.

# Types of Research

We ask questions in three broad ways:

## **Explanatory:**

Why / how did that happen? → Investigate causes, mechanisms

Example: “Why do some democracies collapse?”

- This is the type students often find most interesting but also hardest. It asks "why" or "how."
- Use the example, note this is the kind of question political scientists spend careers trying to answer.
- Also emphasize that explanatory research requires more than just data — it requires a theory about why things happen.

# Types of Research

What type of research is this?

Descriptive: What is happening? → Describe facts, trends, frequencies

Predictive: What will happen? → Forecast outcomes

Explanatory: Why / how did that happen? → Investigate causes, mechanisms

“Will voter turnout increase if polling stations open on weekends?”

- Pause and ask students to classify this question.
- Give students 15–30 seconds to think or discuss with a neighbor before revealing.
- Answer: This is predictive (it asks what will happen under a hypothetical condition).

# Types of Research

What type of research is this?

Descriptive: What is happening? → Describe facts, trends, frequencies

Predictive: What will happen? → Forecast outcomes

Explanatory: Why / how did that happen? → Investigate causes, mechanisms

“What percent of countries held national elections during the COVID-19 pandemic?”

- Pause and ask students again.
- Answer: This is descriptive — it asks for a factual count/percentage.

# Types of Research

What type of research is this?

Descriptive: What is happening? → Describe facts, trends, frequencies

Predictive: What will happen? → Forecast outcomes

Explanatory: Why / how did that happen? → Investigate causes, mechanisms

“Why did support for democracy decline among young voters after 2020?”

- Pause and ask.
- Answer: This is explanatory — it asks for causes or mechanisms behind a trend.
- Wrap up the three quiz slides by noting: "The type of question you ask shapes your research design — the data you collect, the methods you use, and the conclusions you can draw."

# Two Dimensions of Research

## Research Method

|                                        |                                            |
|----------------------------------------|--------------------------------------------|
| What kind of question are we asking?   | What kind of data and analysis do we use?  |
| Descriptive   Predictive   Explanatory | Qualitative   Quantitative   Mixed Methods |
| Goal                                   | Data and methods                           |

- This slide bridges two separate distinctions that students often conflate. The three research types (descriptive, predictive, explanatory) come from Toshkov's taxonomy of empirical research goals. The qualitative/quantitative distinction is about data and methods — a separate choice.
- Key point: a descriptive question can be answered with qualitative or quantitative data. An explanatory question can use interviews or regression. These are independent dimensions, not a single pipeline.
- A common student misconception is thinking descriptive = quantitative or explanatory = qualitative. This slide is designed to prevent that.

The line between **qualitative** and **quantitative** research is increasingly blurred — many projects use both.

- Transition: Type of Research
- This slide makes an important nuance: the boundary between qual and quant is not as sharp as students often think.
- Emphasize the two guiding questions: "What type of data are we using?" and "What kind of analysis are we doing?" These help students navigate the distinction without getting stuck in rigid categories.

The line between qualitative and quantitative research is increasingly blurred  
— many projects use both.

But if we ask:

What type of data are we using?

and

What kind of analysis are we doing?

— we can still understand the key differences and choose the right  
approach for our question.

- In the pilot, the types of data section was one of the areas students found most confusing. Slow down here if needed.

# Types of data

Identify patterns, trends, and relationships

Surveys (e.g., Public opinion polls)

Election Data (e.g., Voter turnout, party vote share)

Index Scores (e.g., Democracy Index, Corruption Perceptions Index)

Government & Economic Data (e.g., GDP, military spending, trade policies)

- Pick one or two examples students have already encountered in the course and connect them. For example: "Remember when we looked at V-Dem democracy scores? That's an index score."
- Emphasize that these are all structured, numeric data — they allow you to identify patterns across many cases.

# Types of data

Identify patterns, trends, and relationships

Surveys, Election Data, Index Scores, Government & Economic Data...

Also

Interviews & Focus Groups (e.g., In-depth conversations with voters about why they support a candidate)

Field Notes / Ethnographic Observation (e.g., Observing a town hall meeting, a protest)

- Point out the contrast: quantitative data gives breadth (many cases), qualitative data gives depth (rich detail on a few cases).
- Ask students: "Can you think of a political question that would be better answered with interviews than with survey data?" This helps them see why both matter.

# Types of data

Archival Records ? Policy Documents & Speeches? Media & Content Analysis?

## Normalization of Censorship: Evidence from China

Tony Zirui Yang

Abstract

Full Text

PDF

Supplemental Material

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**Abstract**

Authoritarian censorship is traditionally understood as a repressive tool and will cause public backlash against the regime. However, surveys worldwide consistently find that citizens in authoritarian regimes are apathetic toward or even supportive of government censorship. To explain this puzzle, I theorize that citizens are subject to a normalization process. Specifically, individuals become desensitized to censorship activities when the range of censored content expands beyond politically threatening topics, like government criticism and collective action, to other less political issues. Using 28 million censored articles and two original survey experiments in China, I show that (1) a majority of censored articles are indeed unrelated to politically threatening topics, and (2) respondents exposed to the censorship of both political and nonpolitical content report higher support for censorship and the regime than those exposed only to political censorship. These findings highlight how normalizing repressive apparatuses like censorship contributes to authoritarian control and survival.

- This slide introduces a gray area: documents, speeches, and media content can be analyzed either qualitatively or quantitatively.
- The screenshot shows Yang (2025). This was chosen because it connects to POLI 101 course content on authoritarianism. Swap this with any article relevant to your course that uses text-as-data methods.

Although documents and texts are traditionally used in qualitative research, they can also be analyzed quantitatively using methods like text analysis or content coding.

- This is the first part of a build explaining how qualitative-seeming data can be turned into quantitative analysis.
- Key message: documents and texts are traditionally qualitative, but methods like text analysis and content coding can make them quantitative.

Although documents and texts are traditionally used in qualitative research, they can also be analyzed quantitatively using methods like text analysis or content coding.

For example, we can:

Count how often certain words or phrases appear in political speeches or news articles

Measure sentiment (positive/negative tone) in social media posts

Track frequency of themes across hundreds of documents over time

This turns **words into data**, allowing us to detect patterns, compare across cases, and run statistical tests — all using quantitative tools.

- Walk through the three examples: counting word frequency, measuring sentiment, tracking themes over time.
- Tip: If students look confused, offer a concrete example: "Imagine counting how many times a president says 'security' vs. 'economy' in their State of the Union addresses over 20 years. That turns speeches into data."
- Mention Unit 3 will introduce AI tools that can help with tasks like this.

# Types of Analysis

Qualitative : Discourse analysis, Case studies/ process tracing, Narrative analysis...

Quantitative : Descriptive statistics, Correlation analysis , Regression analysis...

Many political science projects combine methods: Mixed Methods

The method you choose should align with your **research question**.

- Explain Mixed methods strategically integrates or combines quantitative and qualitative research methods to draw on the strengths of each.
- Key takeaway: "The method you choose should align with your research question."

| Research Question                                                      | Data Type                                            | Analysis Type                                                            |
|------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------|
| 1. How often do first-generation college students attend office hours? | Survey data on office hour attendance & demographics | Descriptive statistics (frequencies, means) – e.g. t-test between groups |
|                                                                        |                                                      |                                                                          |
|                                                                        |                                                      |                                                                          |

Which part is qualitative, which is quantitative?

- Walk through each row and ask students to notice the alignment: the question dictates the data type, which dictates the analysis.

| Research Question                                                      | Data Type                                                    | Analysis Type                                                            |
|------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------|
| 1. How often do first-generation college students attend office hours? | Survey data on office hour attendance & demographics         | Descriptive statistics (frequencies, means) – e.g. t-test between groups |
| 2. Why did youth voter turnout rise in the 2020 U.S. election?         | Interview transcripts with young voters, campaign organizers | Thematic coding to identify motivations, narratives, influences          |
|                                                                        |                                                              |                                                                          |

Which part is qualitative, which is quantitative?

| Research Question                                                                                      | Data Type                                                               | Analysis Type                                                                             |
|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1. How often do first-generation college students attend office hours?                                 | Survey data on office hour attendance & demographics                    | Descriptive statistics (frequencies, means) – e.g. t-test between groups                  |
| 2. Why did youth voter turnout rise in the 2020 U.S. election?                                         | Interview transcripts with young voters, campaign organizers            | Thematic coding to identify motivations, narratives, influences                           |
| 3. How do local politicians talk about immigration in media, and how does it affect voters' attitudes? | Speeches, social media posts (text); pre/post survey on voter attitudes | Text analysis (e.g. sentiment, topic modeling) + regression to test media exposure effect |

Which part is qualitative, which is quantitative?

# Strengths & Limitations

| Aspect      | Qualitative                                                                      |  |
|-------------|----------------------------------------------------------------------------------|--|
| Strengths   | Deep understanding, context, meaning<br><br>Reveals mechanisms & nuance          |  |
| Limitations | Hard to generalize, time/resource intensive<br><br>Small sample size, subjective |  |
| Useful for  | Case-specific questions                                                          |  |

**Each method has trade-offs. Choose based on your research goals.**

- Walk through the qualitative column. Emphasize: deep understanding, context, mechanisms — but hard to generalize, time-intensive, small samples.

# Strengths & Limitations

| Aspect      | Qualitative                                 | Quantitative                                   |
|-------------|---------------------------------------------|------------------------------------------------|
| Strengths   | Deep understanding, context, meaning        | Breadth, generalizability, statistical testing |
|             | Reveals mechanisms & nuance                 | Can identify patterns across many cases        |
| Limitations | Hard to generalize, time/resource intensive | May miss context or meaning                    |
|             | Small sample size, subjective               | Dependent on good data quality                 |
| Useful for  | Case-specific questions                     | Broad comparisons                              |

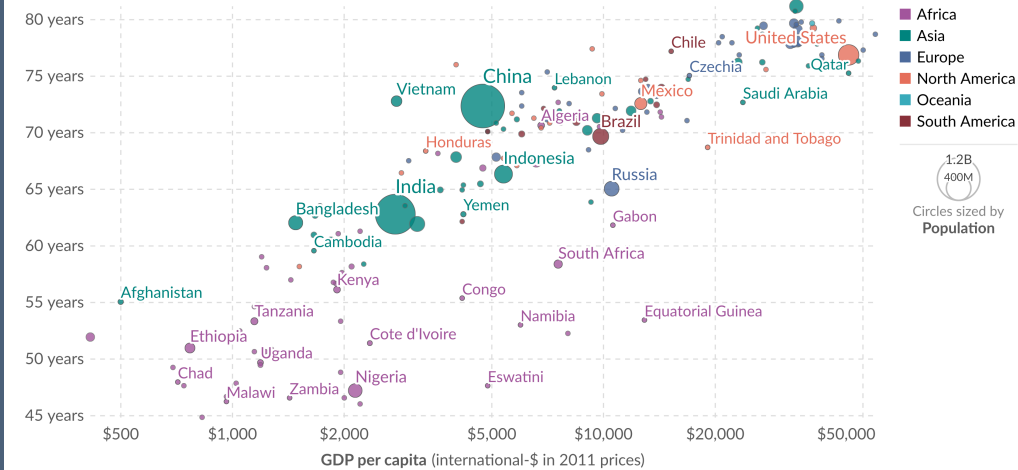
**Each method has trade-offs. Choose based on your research goals.**

- Walk through the quantitative column: breadth, generalizability, statistical power — but may miss context, dependent on data quality.
- End with: "Each method has trade-offs. Choose based on your research goals — there's no single 'best' method."
- Summary moment before transitioning to the next section on statistical inference.

## Life expectancy vs. GDP per capita, 2000

The period life expectancy<sup>1</sup> at birth, in a given year. GDP per capita is adjusted for inflation and differences in living costs between countries.

Life expectancy at birth



Higher GDP = Longer life?

- Use this as a visual hook before defining anything. Ask: "What do you see? What's the relationship?"
- Students should notice the positive but curved pattern — countries with higher GDP tend to have longer life expectancy, but the relationship flattens at higher income levels.
- Point out that circle sizes represent population — this is a good moment to highlight that one chart can encode multiple variables at once.
- Source: Our World in Data, "Life expectancy vs. GDP per capita, 2000." <https://ourworldindata.org/grapher/life-expectancy-vs-gdp-per-capita>

# What Is Statistical Inference?

In political science, we often want to ask:

“Is there a meaningful relationship between two variables?”

## **Statistical inference:**

The process of analyzing data generated by a sample, but then used to determine some characteristic of the larger population.

Keep it intuitive: "We can't study every country or every voter. So we use samples and statistics to make informed claims about the bigger picture.”

Context: In the pilot, this entire statistical inference section (Slides 24–34) was not included in exams or other evaluations. Students were assessed on political science methods concepts, but the t-test and p-value content here is a demonstration. The goal is exposure — showing what statistical reasoning looks like in practice, this is a "see how this works" walkthrough, not a "learn how to do this" lesson. Adjust expectations accordingly.

# What Is Statistical Inference?

"Is there a meaningful relationship between two variables?"

Statistical inference helps us test this using sample data.

- Independent Variable (IV): the "cause" (what we change or compare)

- Dependent Variable (DV): the "effect" (what we observe)

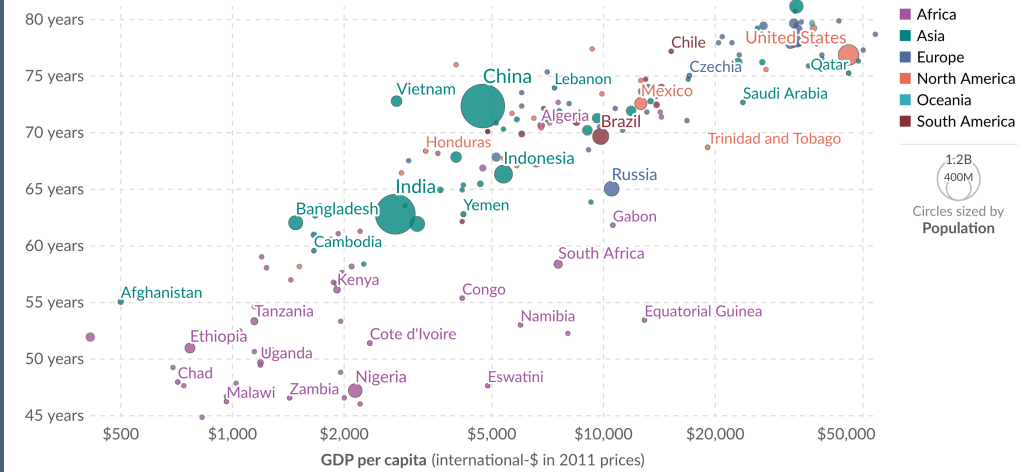
➤ We check if the IV and DV are related using statistical tests.

- Frame the connection: "Statistical tests help us check whether the IV and DV are actually related, or whether the pattern we see is just noise."
- Mention: IV is also called the explanatory variable, predictor, or treatment variable. DV is also called the outcome variable, response variable, or dependent measure. These all refer to the same underlying logic — what we think does the influencing vs. what we think gets influenced.

## Life expectancy vs. GDP per capita, 2000

The period life expectancy<sup>1</sup> at birth, in a given year. GDP per capita is adjusted for inflation and differences in living costs between countries.

Life expectancy at birth



IV (cause)? DV(effect)?

- Same scatterplot returns. Now ask students to apply what they just reviewed: "Which is the IV and which is the DV?"
- Answer: GDP per capita = IV; life expectancy = DV. We're asking whether wealth predicts longer life, not the other way around.
- If students struggle, prompt: "Which variable do we think might influence the other?"

## Key concepts:

Independent Variable (IV): the “cause”

Dependent Variable (DV): the “effect”

Correlation = two variables move together

Causation = one variable *produces* change in another

► Just because two things are correlated doesn't mean one caused the other!



- Review: correlation = two variables move together; causation = one actually produces change in another.
- Meme: (made by student from prev class) someone mistakenly thinking they are “moving the truck” (correlation) versus the person actually pushing it (causation).
- Real-world example: ice cream sales and drowning rates are correlated — but ice cream doesn't cause drowning. The hidden variable is summer/heat.
- Optional reference: Spurious Correlations website (<https://tylervigen.com>)

## Example: T-test

Do students who attend office hours score higher on the final exam?

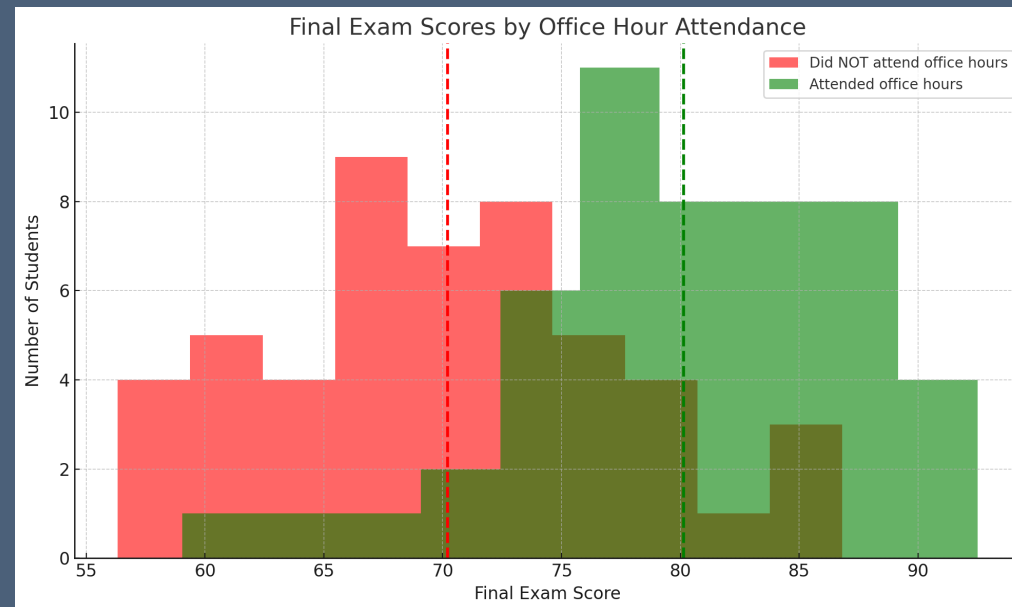
IV: Attending office hours (Yes/No)

DV: Final exam score

Method: Compare the mean (exam score) of two groups?

- Introduce the scenario
- Walk through the setup
- Ask: "What do you think — does going to office hours actually help?" Let a few students respond before continuing.
- Remind them: we're not trying to teach them how to run a t-test. This is about understanding the logic of how researchers test claims.

Do students who attend office hours score higher on the final exam?



- The histogram visually shows two overlapping distributions — red (did not attend) and green (attended).
- Point out the dashed lines marking each group's mean. The green group is shifted to the right.
- Key question: "The means look different — but how do we know it's a real difference and not just random variation?"

## Example: T-test

Do students who attend office hours score higher on the final exam?

IV: Attending office hours (Yes/No)

DV: Final exam score

Method: Compare the mean (exam score) of two groups?

A t-test is a statistical test used to compare means between two groups. It helps determine whether the difference between the groups is statistically significant or just due to chance.

- Keep it conceptual — no need for formulas or calculations.
- Analogy if helpful: "Imagine flipping a coin 100 times and getting 70 heads. Is the coin unfair, or did you just get lucky? A t-test helps answer that kind of question."

## Example: T-test

Do students who attend office hours score higher on the final exam?

IV: Attending office hours (Yes/No)

DV: Final exam score

Method: T test. Compare the mean (exam scores) of two groups.

| Group                       | N  | Mean Score | Std. Dev. |
|-----------------------------|----|------------|-----------|
| Did NOT attend office hours | 50 | 72.1       | 7.9       |
| Attended office hours       | 50 | 80.3       | 8.1       |

- Show the numbers: non-attenders averaged 72.1 (SD = 7.9); attenders averaged 80.3 (SD = 8.1). N = 50 per group.
- Ask: "That's about an 8-point gap. Sounds meaningful — but is it? How do we know it's not just luck?"
- The standard deviation tells us how spread out each group's scores are. If scores are very spread out, even a big mean difference might not be reliable.
- Transition: "This is where the p-value comes in."

Do students who attend office hours score higher on the final exam?

IV: Attending office hours (Yes/No)

DV: Final exam score

**P- Value:**

A number (between 0 and 1) that tells us how likely our results are due to random chance

Common threshold:  $p < 0.05$  = less than 5% chance it happened randomly

| Statistic               | Value                                            |
|-------------------------|--------------------------------------------------|
| t-statistic             | 5.18                                             |
| <b>p-value</b>          | 0.00001                                          |
| df (degrees of freedom) | 98                                               |
| Interpretation          | $p < 0.05 \rightarrow$ statistically significant |

- Define p-value: a number between 0 and 1 that tells us how likely our results are if there were actually no real difference (i.e., due to random chance alone).

- Explain if needed:

- t-statistic

A number that tells you how big the difference is between two group averages, compared to how much the scores vary. Bigger = stronger difference.

- Degrees of Freedom (df)

A technical value that's kind of like sample size minus 2. It helps the computer figure out what counts as a "big" t-statistic.

- p-value

A number that says: "What are the odds we'd see this difference by random chance?" A small p-value (like 0.03) means "This is probably a real difference."

## Do students who attend office hours score higher on the final exam?

Common threshold:  $p < 0.05$  = less than 5% chance it happened randomly

Smaller p-value = more evidence that there's a real relationship

➤ If  $p < 0.05$  → we might say office hours *make a difference*

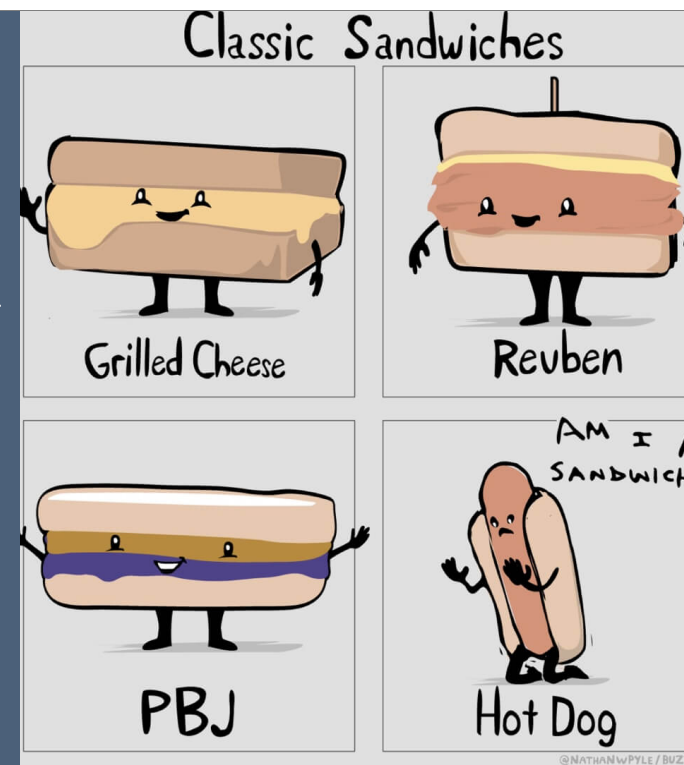
➤ If  $p > 0.05$  → difference could just be random

| Statistic               | Value                                             |
|-------------------------|---------------------------------------------------|
| t-statistic             | 5.18                                              |
| <b>p-value</b>          | 0.00001                                           |
| df (degrees of freedom) | 98                                                |
| Interpretation          | $p < 0.05$ → statistically significant difference |

- Walk through both scenarios on the slide:  $p < 0.05$  → we'd say office hours make a difference;  $p > 0.05$  → the difference could just be random.
- Important caveat: "Even with a tiny p-value, this doesn't prove office hours cause higher scores. Students who go to office hours might just be more motivated to begin with. Remember: correlation  $\neq$  causation."
- Ask "So what else could we do to figure out whether office hours actually help?" Prompt for ideas — e.g., interviewing both groups to understand their study habits, running an experiment where students are randomly assigned to attend. This connects back to the qual vs. quant discussion and shows why researchers often need multiple methods.

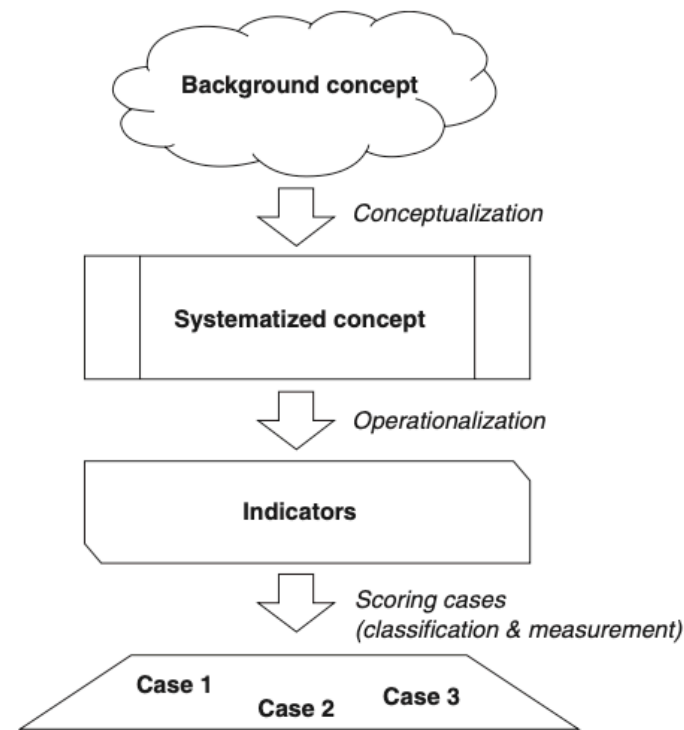
# Conceptualization to Measurement

V Dem EDI



- Transition: "We've talked about types of questions, data, and analysis. Now let's zoom into one of the hardest parts of research — turning abstract ideas into something we can actually measure."
- Ask: "Is a hot dog a sandwich?" Let students react.
- Then make the point: "If we can't even agree on what counts as a sandwich, imagine how hard it is to define 'democracy.' And the definition matters — if I ask 'How many sandwiches were sold at the SC State Fair?' my answer changes entirely depending on whether I count hot dogs or not. Same question, different definition, different answer. That's exactly what happens in political science when researchers define concepts differently."

- Democracy



- This diagram is adapted from Adcock & Collier (2001), as presented in Toshkov, Ch. 4.
- Walk through the four levels top to bottom: background concept (the broad idea) → conceptualization (defining it precisely) → operationalization (choosing indicators) → scoring cases (assigning values).
- Students have seen these terms in POLI 101 before but haven't applied them to a real dataset. This section makes it concrete using V-Dem's Electoral Democracy Index.
- (Optional) The concept-to-measurement walkthrough uses democracy and V-Dem as the running example. If you prefer a different concept, we recommend choosing one that maps to variables in the POLI-LIT teaching dataset. This way the example connects directly to what students will work with in Units 1–3. The pipeline structure still works — just swap the concept, indicators, and coding at each step.

## From Concept → Measurement (Process)

To study an abstract idea (like democracy, legitimacy, representation), we follow this pipeline:

1. Conceptualization: What *does* the idea mean?
2. Indicator(s): What observable things will represent that concept?
3. Measurement / Coding: How do we assign values (scores, categories) to those indicators?

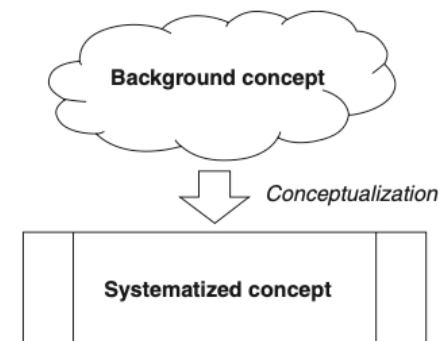
- Context: In the pilot, students had already encountered Dahl's two dimensions of democracy (contestatory and inclusionary) earlier in the course. This section serves as a review and application of that framework to a real dataset (V-Dem EDI). If your class has not covered Dahl, you'll need to teach the two dimensions here rather than just review them or swap the concept.

# Conceptualization

A concept is a mental or theoretical notion (e.g., democracy, freedom, representation).

Conceptualization is the process of defining exactly what we mean by that concept in our study.

**Figure 4.2** *Conceptualization and operationalization*



- Example (democracy): "A system where rulers are held accountable through elections, citizens can speak freely, and public decisions rest with elected officials."

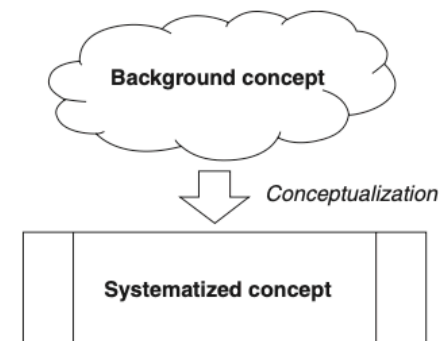
# Conceptualization

A concept is a mental or theoretical notion (e.g., democracy, freedom, representation).

Conceptualization is the process of defining exactly what we mean by that concept in our study.

The process of creating mental categories that capture the meaning of objects, events, or ideas.

**Figure 4.2** *Conceptualization and operationalization*



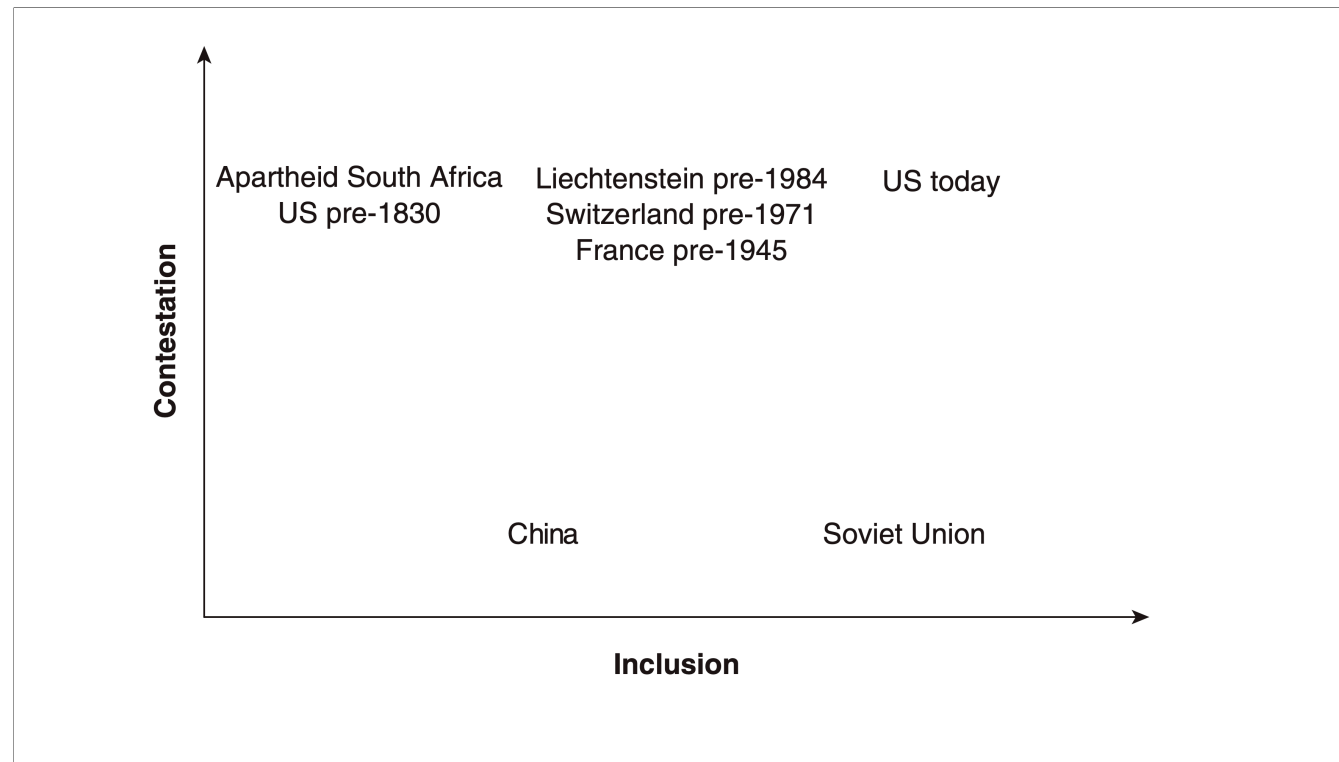
## Conceptualization: What *does* democracy mean?

Robert Dahl proposed a minimalist view of democracy.

Two dimensions:

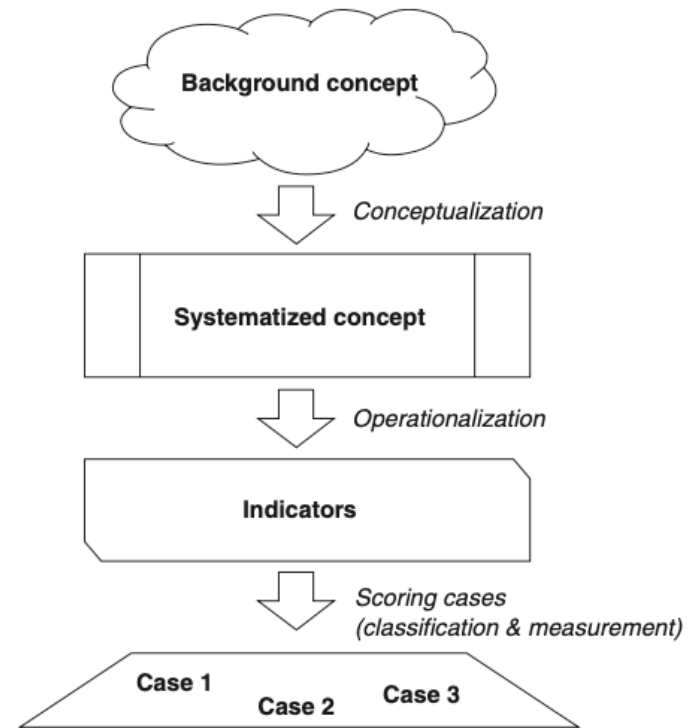
1. Contestation captures the extent to which citizens are free to organize themselves into competing blocs in order to press for the policies and outcomes they desire.
2. Inclusion has to do with who gets to participate in the democratic process.

- This is one of the most influential conceptualizations in the field. Note that it's deliberately narrow — it doesn't include things like economic equality or press freedom.
- Ask: "Do you think this captures everything we mean by democracy? What's missing?"



- This plot maps real-world cases onto Dahl's two dimensions.
- Walk through a few examples: US today = high on both; China = low on both; Apartheid South Africa = high contestation but low inclusion; Soviet Union = high inclusion (universal suffrage) but low contestation.
- Students often find the historical examples surprising (e.g., Switzerland pre-1971 didn't allow women to vote → high contestation, limited inclusion).
- Source: Adapted from Clark, William R., Matt Golder, and Sona N. Golder. Principles of Comparative Politics, 4th Edition, Ch. 5 ("Democracy and Dictatorship: Conceptualization and Measurement").

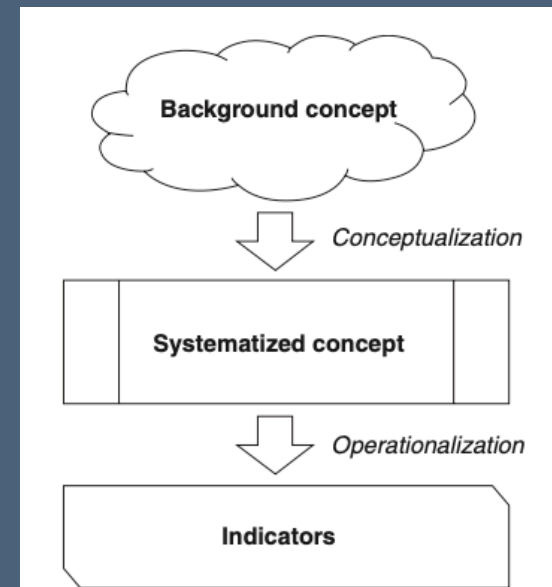
- Democracy
- Contestation and Participation/Inclusion



- Diagram now shows "Democracy" as background concept and "Contestation and Participation/Inclusion" as the systematized concept.
- Reinforce: "We've now defined what we mean by democracy. Next: how do we observe it?"

## Operationalization — Choosing Indicators

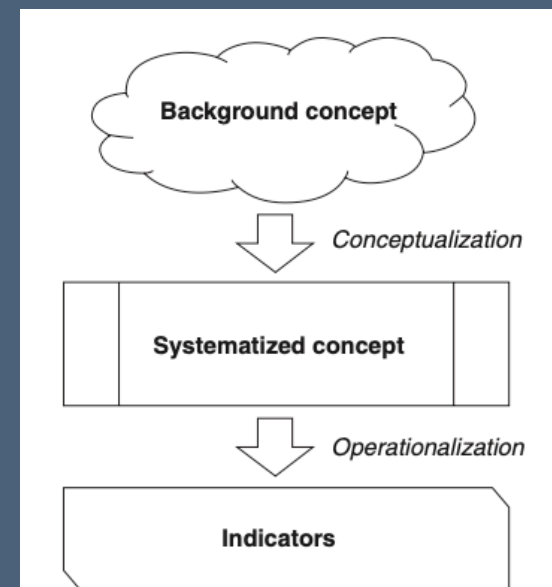
Indicators are observable phenomena that stand in for the abstract concept.



## Operationalization — Choosing Indicators

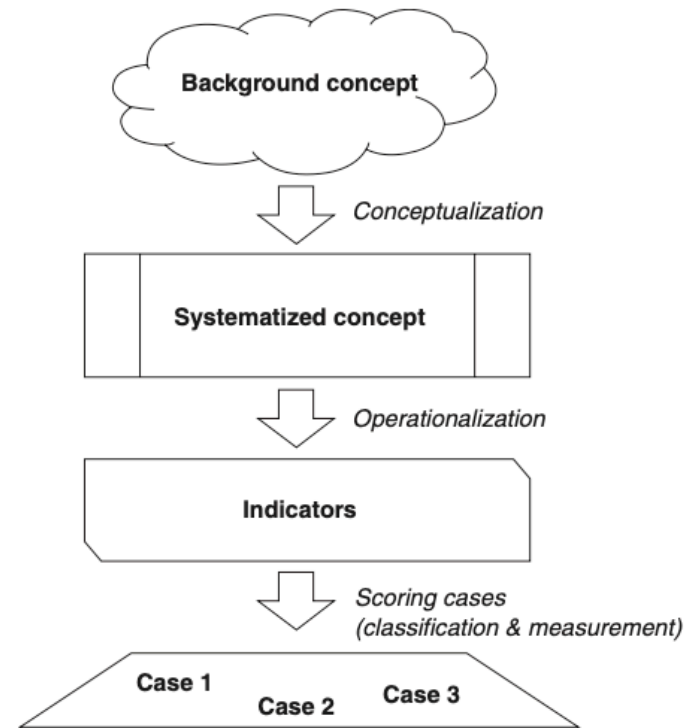
For Dahl's concept of democracy, possible indicators include:

- "Were elections free & fair?"
- "Freedom of speech / media"
- "Right of association (political parties, civic groups)"
- "Extent of suffrage / inclusion"



- For Dahl's democracy: free & fair elections, freedom of speech/media, right of association, extent of suffrage.
- Each of these is something a researcher could actually observe or measure in the real world — that's what makes them indicators.
- Ask: "Can you think of other indicators someone might use for democracy that aren't on this list?"

- Democracy
- Contestation and Participation/Inclusion
- Regular election? Voting Right? Effective party counts?

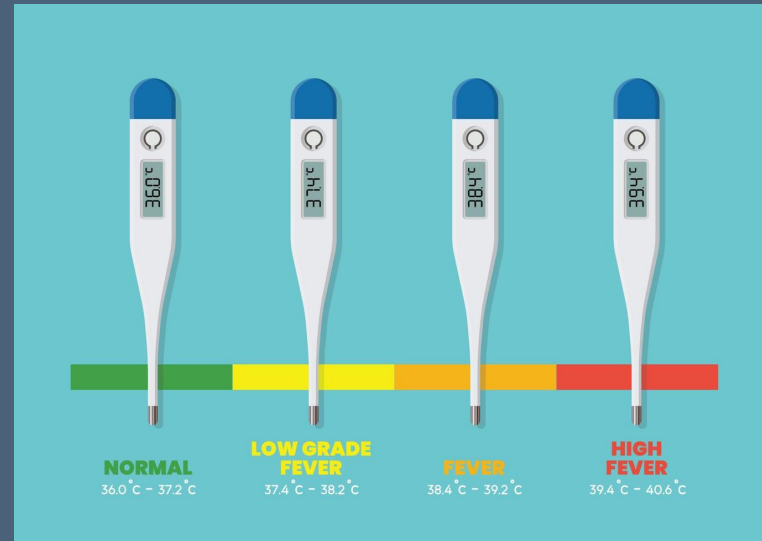


- Transition: "Now we know what to look for. But how do we actually assign numbers or scores?"

|                                  |                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Varieties of Democracy           | <ul style="list-style-type: none"> <li>Narrow and broader: electoral, liberal, participatory, deliberative, or egalitarian democracy</li> </ul> |
| Regimes of the World             | <ul style="list-style-type: none"> <li>Narrow: electoral or liberal democracy</li> </ul>                                                        |
| Lexical Index                    | <ul style="list-style-type: none"> <li>Narrow: electoral (or liberal) democracy</li> </ul>                                                      |
| Boix-Miller-Rosato               | <ul style="list-style-type: none"> <li>Narrow: electoral democracy</li> </ul>                                                                   |
| Polity                           | <ul style="list-style-type: none"> <li>Narrow: electoral and liberal democracy</li> </ul>                                                       |
| Freedom House                    | <ul style="list-style-type: none"> <li>Narrow: electoral or liberal democracy</li> </ul>                                                        |
| Bertelsmann Transformation Index | <ul style="list-style-type: none"> <li>Broad: electoral, liberal, participatory, deliberative, and effective democracy</li> </ul>               |
| Economist Intelligence Unit      | <ul style="list-style-type: none"> <li>Broad: electoral, liberal, participatory, deliberative, and effective democracy</li> </ul>               |

- This table shows how different projects (V-Dem, Regimes of the World, Freedom House, Polity, etc.) conceptualize democracy differently — some narrow (electoral only), some broad (electoral + liberal + participatory + deliberative).
- Key takeaway: the same concept can be measured in very different ways depending on the choices researchers make. There is no single "right" democracy index.

## Measurement / Coding



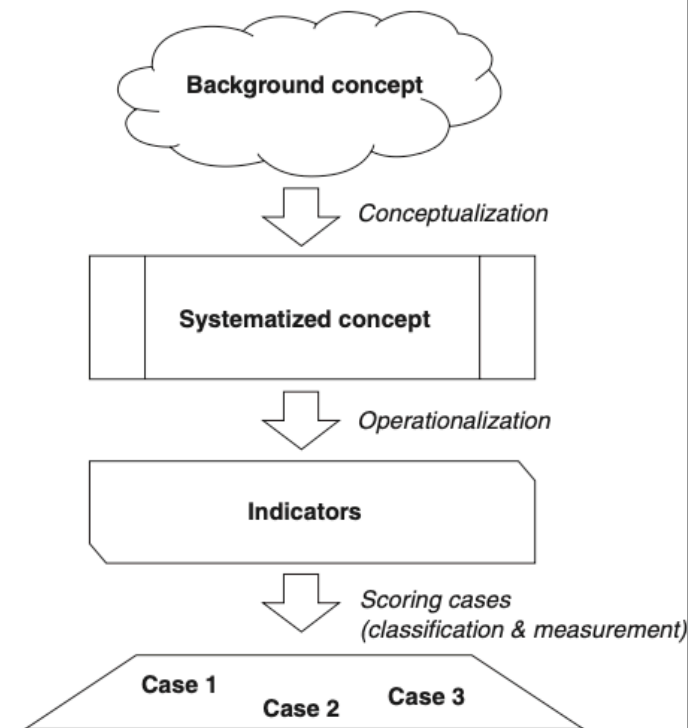
- Analogy: just as temperature is measured and categorized (normal, low fever, fever, high fever), political concepts get coded into values too.

## Measurement / Coding

Once indicators are chosen, we must measure or code them:

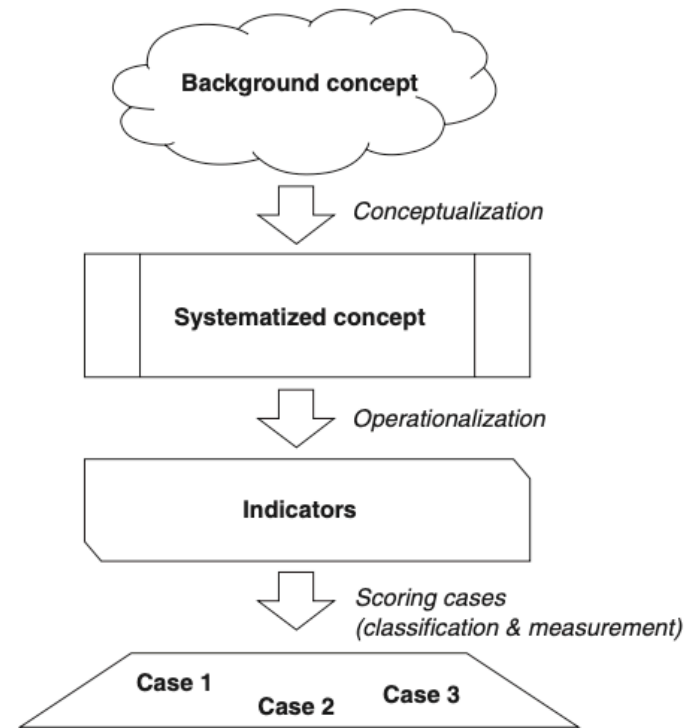
- Surveys (e.g. Likert scales)
- Expert coding (e.g. ordinal ratings)
- Event counts / official data

The coding gives us numeric or categorical values for each indicator, which can then be aggregated.



- Once indicators are chosen, researchers must code them: surveys (Likert scales), expert coding (ordinal ratings), event counts / official data.
- The coding produces numeric or categorical values for each indicator, which can then be aggregated into an index.

- Democracy
- Contestation and Inclusion
- Regular election? Voting Right?  
Effective party counts?
- Democracy Index



- Full pipeline revealed: Democracy → Contestation & Inclusion → Elections, Voting Rights, Party Counts → Democracy Index.
- Pause: "This is the whole journey from an abstract idea to a number in a dataset. A lot of numbers you see in democracy index went through a process like this."

## Sample V-Dem Coding / Survey Questions

Here are real items from the **V-Dem v15 codebook**:

- “To what extent are national elections free and fair?” (ordinal expert rating)
- “To what extent is there freedom of expression and alternative sources of media?”
- “To what extent do citizens have the right to associate in political parties or civil society organizations?”
- “To what degree do public policy decisions lie with elected officials rather than unelected actors?”

- Context: Like the t-test section, this part is a demonstration — students don't need to understand the technical details of Bayesian models or aggregation formulas. The goal is for them to see that building an index involves many choices, and those choices matter.

# Now what?



- Brief pause/transition slide. The question is: once we have all these coded responses (tons of survey result from different country/case experts), what do we do with them?

# Aggregation & Index Construction (V-Dem)

After coding, V-Dem builds indices via:

1. Grouping indicators into subcomponents (e.g. election quality, civil liberties)
2. Using a latent variable / Bayesian measurement model to produce continuous estimates, accounting for coder differences, uncertainty, and bias

- Flag key issues at this stage:
- Subjectivity / coder bias: expert judgments can vary across coders and over time.
- Missing data or weak indicators: some countries lack full coverage, especially in earlier years or for smaller states.
- Aggregation choices / weights: how much weight each subindicator gets affects the final score. Different weighting = different results.

# Aggregation & Index Construction (V-Dem)

After coding, V-Dem builds indices via:

1. Grouping indicators into subcomponents (e.g. election quality, civil liberties)
2. Using a latent variable / Bayesian measurement model to produce continuous estimates, accounting for coder differences, uncertainty, and bias
3. Combining subindices into high-level indices, like the Electoral Democracy Index (EDI / "v2x\_polyarchy")
4. Producing point estimates and uncertainty intervals for each country-year

Interpreting & Critiquing the Index?

- Flag additional issues:
- Uncertainty: every V-Dem score is an estimate, not a precise truth. V-Dem provides credible intervals — a range that says "the true value is probably somewhere in here." That's a sign of good measurement practice.
- Version comparability: scores from different V-Dem releases may shift due to updated coding rounds or calibration changes. Always check which version you're using.
- Bottom line for students: always check the codebook, methodology documentation, and uncertainty bounds before interpreting results. This applies to V-Dem and to any index in political science.
- Transition: "So how do we interpret — and critique — the final index?"

The process we traced:

Background Concept (Democracy)

→ Conceptualization — capture the meaning (Contestation and Inclusion)

→ Operationalization — choosing Indicators (free elections, association, media freedom, etc.)

- Recap the process

The process we traced:

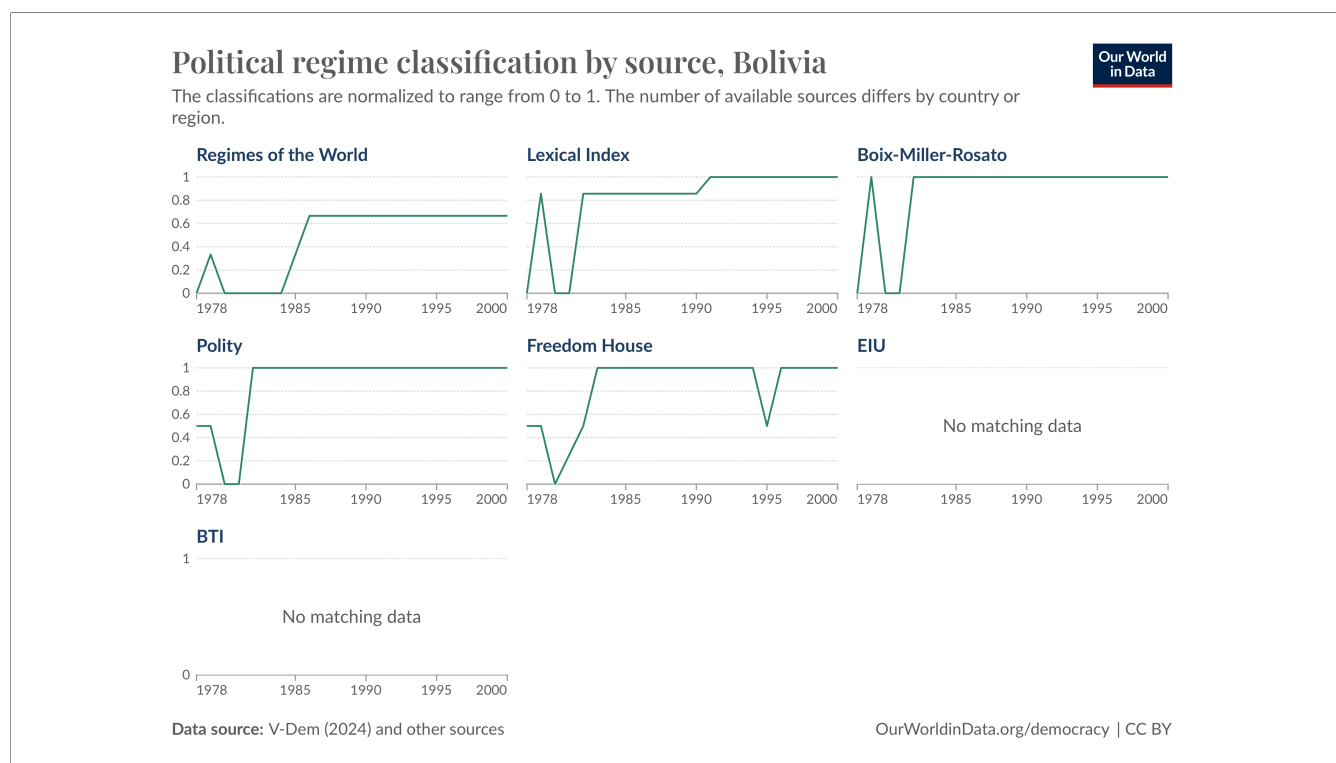
Background Concept (Democracy)

- Conceptualization — capture the meaning (Contestation and Inclusion)
- Operationalization — choosing Indicators (free elections, association, media freedom, etc.)
- Measurement — coding (expert ratings, ordinal scores)

Aggregation / Index (V-Dem EDI)

This example helps you see concretely how decisions at each step shape the final measure.

- Key message: "Decisions at every step shape the final number. If you change the definition, the indicators, or the coding method, you could get a different score for the same country in the same year."
- This ties back to the earlier slide about different democracy indices.



- This Our World in Data chart shows how different indices classify Bolivia's regime over time. Some show gradual change, others show sharp jumps; some have data, others don't.
- Use this as the punchline: "Same country, same time period — different answers depending on which index you use. That's why understanding how data is constructed matters."
- Source: Our World in Data, "Political regime classification by source." <https://ourworldindata.org/grapher/political-regime-classification-by-source>

# POLI LIT Unit 1

## Preview

POLI-LIT = 1 in-person lecture (today) + 3 online self-paced units

**This Wednesday** → Unit 1 opens online:

- Learn about the kinds of data used in political science
- What they look like and where they come from
- First steps before using data
- Basic statistical ideas (like “average” and “distribution”)

- Transition: "Now you've seen how political scientists turn ideas into data. Starting this week, you'll get hands-on practice with data yourself.”
- Explain that today is the only in-person session — the rest is online and self-paced.
- Unit 1 is all about helping students get comfortable with real political data — no software or math knowledge needed yet.
- Clarify the module structure: 1 in-person lecture (today) + 3 online units.
- Context: In the pilot (Fall 2025, POLI 101), the module originally had only Units 1 and 2. After piloting, we found some sections were too dense, so we restructured the content into three units (Units 1–3) to improve pacing. The current three-unit structure reflects that revision. Feel free to adjust timeline/workload based on students' needs.

# POLI LIT Unit 1

## Preview

Later in the semester (**Unit 2**):

- Intro to **R** (no download needed)
- Use **AI tools** to help with coding & data visualization

Each unit includes practice questions + open-ended reflection prompts

- Emphasize that the module is designed for beginners. Unit 2 & 3 (later in the semester) will walk students into coding and visualization using R and AI — but gently. No need to worry about it now.
- Each unit includes practice questions + open-ended reflection prompts. Encourage students to take both seriously — the reflection prompts help them think critically; the multiple-choice questions help the instructors see how students are doing.
- All units are completion-based — no penalty as long as they complete it.

# POLI LIT Unit 1

## Preview

Later in the semester (**Unit 2**):

- Intro to **R** (no download needed)
- Use **AI tools** to help with coding & data visualization

Each unit includes practice questions + open-ended reflection prompts

**Complete the pre-course survey = earn 1 extra credit**

**Unit 1 = 5% of your course grade (completion-based) , ~ 1 hr**

- Pilot logistics: Unit 1 = 5% of course grade (completion-based, ~1 hour). Pre-course survey = 1 extra credit point. No penalty as long as students complete the work.
- Context: adjust grading weight, deadlines, and extra credit structure to fit your course. The completion-based approach worked well in the pilot — it kept stakes low and encouraged genuine engagement rather than performance anxiety.
- If you use a pre/post survey for assessment or research purposes, this is a good moment to mention it and share the link.

### Realistic political data

- We created a mock dataset + codebook based on real sources
- You'll explore how data is used in research

### Unit 1 covers:

- What political data looks like
- How to read a **codebook**
- What to check before you use data

Unit 1 (Oct 8)  
Visible to students

Introduction  
Visible to students

Understanding Data  
Visible to students

Practice: Reading Codebook  
Visible to students

Basic Statistical Concepts  
Visible to students

Ethical use of data  
Visible to students

Unit 1 Summary  
Visible to students

Unit 1 Feedback  
Visible to students  
Due date: 10/10/25, 11:59 PM (EDT)

- Let students know that POLI-LIT uses real-world-style political data, but it's been simplified into a teaching dataset + codebook to make it approachable.
- Reassure them: no R coding yet in Unit 1. That starts in Unit 2. This unit is about getting familiar with how political data is structured and interpreted.
- Highlight that they'll learn what a codebook is — a tool researchers use to understand variables — and how to spot issues in a dataset before trying to analyze it.
- Connect back to today's lecture: "Everything we talked about today — conceptualization, measurement, coding — you'll see that play out in the dataset and codebook."

**Basic statistical concepts:**

- What is a mean, median, range?
- What does a “distribution” tell us?
- First steps in telling stories with data

**Ethical thinking with data:**

- How is data produced and collected?
- Who decides what gets measured and why?

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- The statistics content in Unit 1 is very introductory — mean, median, range, distributions. Focus on basic terms and data shapes. Not math-heavy.
- The ethical thinking section asks students to consider how data is produced, who decides what gets measured, and why those choices matter. This connects directly to the conceptualization-to-measurement pipeline from today's lecture.
- End with a reminder: Unit 1 closes with a required feedback form. It's short, and it will help you (the instructors) make this module better for everyone.
- Pilot logistics: the feedback form had a set due date (completion-based).
- Suggested closing line: "You don't need to be a data expert to do well in POLI-LIT. You just need to be curious and willing to try. We designed this to meet you where you are — so take your time, ask questions, and have fun with it."

# Update

- Take Home Assignment #1 Due today
- Return your survey and sign the sheet!
- Unit 1 (online) - Complete between Wed - Fri
- Exam Study Guide - Wed